Xi (Chase) **Jiang**

CS PH.D., UCHICAGO; CO-FOUNDER, ARES

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Summary

Now working on neural AI at a cool company. Prev. PhD in Computer Science at the University of Chicago, with research centered on applying AI to complex, high-dimensional data — spanning synthetic data generation, foundation model pretraining, state space modeling, and efficient real-time inference. Work includes ML-driven data modeling, protocol-constrained data generation, generative modeling for realistic high-complexity data, and real-time data systems, with publications in SIGMETRICS, KDD, WWW, and CoNEXT.

Education

The University of Chicago — *Chicago, IL*

PH.D. IN COMPUTER SCIENCE — *Sep. 2021 - Aug. 2026*

- Advisor: Prof. Nick Feamster (ACM Fellow, MIT Tech Review 35 under 35)
- Recipient of Crerar Fellowship

Colgate University — *Hamilton, NY*

B.S. IN COMPUTER SCIENCE AND ECONOMICS — *Sep. 2017 - May. 2021*

- High Honors in Computer Science
- Laura Sanchis Award for Excellence in Research

Selected Experience

Ares — *New York, NY*

CO-FOUNDER, CTO — *Jul. 2025 - Present*

- Co-founded Ares, a neuro AI company.

NetMicroscope — *Chicago, IL*

MACHINE LEARNING ENGINEER — *May. 2025 - Sep. 2025*

- Developed ML algorithms for anomaly detection in large-scale enterprise network data, deployed at **Apple** and **Comcast**.

Rockfish AI — *Remote*

MACHINE LEARNING ENGINEER INTERN — *Jun. 2024 - Sep. 2024*

- Developed a patented preprocessing method for generative-AI synthetic data, increasing data capacity by **100x** and training speed by **30x**, deployed across organizations including **AWS, Snowflake, Ford, CMU CyLab, and the U.S. Army**.

Red Pulse — *Shanghai, China*

TIER 1 RESEARCH ANALYST — *May. 2019 - May. 2020*

- Analyzed Chinese technology and financial markets and contributed to industry research on the telecommunications sector.

Yinlang Limited — *Shanghai, China*

SDE INTERN — *May. 2018 - Aug. 2018*

- Built real-time user analytics and applied clustering and predictive modeling to behavioral data for a WeChat mini-program.

China Everbright Bank — *China*

FINANCIAL MODELING INTERN — *Jul. 2017 - Aug. 2017*

- Developed financial models for loan evaluation, incorporating borrower financials, repayment capacity, and credit-risk factors to support lending decisions.

China Construction Bank (Asia) Corporation Limited — *Shenyang, China*

CREDIT RISK MODELING INTERN — *Jun. 2017 - Jul. 2017*

- Developed algorithms for mortgage risk assessment, modeling borrower financial profiles, loan characteristics, and repayment risk to support credit evaluation.

Staubli Robotics — *Greenville, SC*

PROGRAMMING INTERN — *May. 2015 - Aug. 2016*

- Improved industrial robot operating efficiency by approximately **2x** through path-optimization algorithms.

Selected Publications

PhD Thesis

[Thesis] Generative Modeling for Packet-Level Network Traffic Synthesis — *Thesis*

Xi Jiang — *2026*

University of Chicago Dissertations

ML FOR NETWORKS

[In submission] NetJump: Controllable Synthesis of Rare Network Conditions via Positional Phase Discontinuity — *In Submission*

Xi Jiang, Francesco Bronzino, Paul Schmitt, Nick Feamster — *In Submission*

In Submission

[Conference 1] CAP: Detecting Network Device Misconfigurations with Context-Aware Prompting of LLMs — *Ann Arbor, MI*

Xi Jiang, Aaron Gember-Jacobson, Nick Feamster — *June 2026*

ACM Special Interest Group on Measurement and Evaluation (**SIGMETRICS'26)**

[Conference 2] SwiftQueue: Optimizing Low-Latency Applications with Swift Packet Queuing — *Virtual*

Siddhant Ray, **Xi Jiang**, Jack Luo, Nick Feamster, Junchen Jiang — *Feb 2026*

New Ideas in Networked Systems (**NINeS'26)**

[Conference 3] NetSSM: Comprehensive and Realistic Network Trace Generation via State Space Models — *HKUST, Hong Kong*

Andrew Chu*, **Xi Jiang***, Shinan Liu, Arjun Nitin Bhagoji, Paul Schmitt, Francesco Bronzino, and Nick Feamster — *Dec 2025*

International Conference on emerging Networking EXperiments and Technologies (**CoNEXT'25)**

[Conference 4] JITI: Dynamic Model Serving for Just-in-Time Traffic Inference — *HKUST, Hong Kong*

Xi Jiang, Shinan Liu, Saloua Naama, Francesco Bronzino, Paul Schmitt, and Nick Feamster — *Dec 2025*

International Conference on emerging Networking EXperiments and Technologies (**CoNEXT'25)**

[Workshop 1] Feasibility of State Space Models for Network Traffic Generation — *Sydney, Australia*

Andrew Chu*, **Xi Jiang***, Shinan Liu, Arjun Nitin Bhagoji, Paul Schmitt, Francesco Bronzino, and Nick Feamster — *Aug 2024*

ACM SIGCOMM (**SIGCOMM'24 NAIC Workshop)**

[Conference 5] NetDiffusion: Network Data Augmentation Through Protocol-Constrained Traffic Generation — *Venice, Italy*

Xi Jiang, Shinan Liu, Aaron Gember-Jacobson, Arjun Nitin Bhagoji, Paul Schmitt, Francesco Bronzino, and Nick Feamster — *June 2024*

ACM Special Interest Group on Measurement and Evaluation (**SIGMETRICS'24)**; Open-Sourced Repo with 80+ stars and used as field standard benchmark (`https://github.com/noise-lab/NetDiffusion`)

[Poster 1] Transformer-based Predictions for Sudden Network Changes — *SANTA CLARA, CA*

Siddhant Ray, **Xi Jiang**, Zhuohan Gu, Junchen Jiang, and Nick Feamster — *April 2024*

21st USENIX Symposium on Networked Systems Design and Implementation (**NSDI'24 Poster)**

[Workshop 2] Generative, High-Fidelity Network Traces — *Boston, MA*

Xi Jiang*, Shinan Liu*, Aaron Gember-Jacobson, Paul Schmitt, Francesco Bronzino, and Nick Feamster — *Nov. 2023*

ACM Workshop on Hot Topics in Networks (**HotNets'23)**

[Conference 6] Augmenting Rule-based DNS Censorship Detection at Scale with Machine Learning — *Long Beach, CA*

Xi Jiang*, Jacob Brown*, Van Tran*, Arjun Nitin Bhagoji, Nguyen Phong Hoang, Nick Feamster, Prateek Mittal, Vinod Yegneswaran — *Aug. 2023*

ACM SIGKDD Conference on Knowledge Discovery and Data Mining (**KDD'23**)

[Workshop 3] Towards Designing Robust and Efficient Classifiers for Encrypted Traffic in the Modern Internet — *Belgrade, Serbia*

Xi Jiang, Shinan Liu, Saloua Naama, Francesco Bronzino, Paul Schmitt, and Nick Feamster — *Oct. 2022*

IAB workshop on Management Techniques in Encrypted Networks **(IAB M-TEN'22)**

NETWORKS MEASUREMENT AND TESTING

[Journal 1] Automating Internet of Things Network Traffic Collection with Robotic Arm Interactions — *Journal Publication*

Xi Jiang, Noah Apthorpe — *May. 2023*

Journal of Communications 18 (5), 283–293 **(JCM)**

[Conference 7] Measuring and Evading Turkmenistan's Internet Censorship — *Austin, TX*

Sadia Nourin, Van Tran, **Xi Jiang**, Kevin Bock, Nick Feamster, Nguyen Phong Hong, Dave Levin — *April. 2023*

ACM Web Conference **(WWW'23)**

[Poster 2] Non-interoperability Detection for Routing Protocol Implementations — *Virtual*

Xi Jiang, Aaron Gember-Jacobson — *Aug. 2021*

ACM SIGCOMM Poster and Demo **(SIGCOMM'21 Poster)**
First Place in the Undergraduate Student Research Competition

[Workshop 4] A Permutation Representation of Covering Arrays — *Virtual*

Ryan E. Dougherty, **Xi Jiang** — *May. 2021*

Genetic Improvement Workshop at the International Conference on Software Engineering **(GI @ ICSE'21)**

General Large Language Model

[Journal 2] Humanity's Last Exam — *Journal Publication*

Long Phan*, Alice Gatti*,..., **Xi Jiang***, *et al.* — *2026*

Nature, Volume 649, Pages 1139–1146 (2026)

Supporting Research Fundings

ANR Project MINT (ANR-21-CE94-0001),
INTERFERE Project (Fédération Informatique de Lyon),
France–Chicago Center: Collaborating in the Sciences,
DARPA RACE Program,

Selected Patents

2025	**Provisional Patent Filing: ENABLING SCALABLE GENERATIVE MODEL TRAINING FOR HIGH CARDINALITY EVENT STREAMS USING STATE AWARE BINNING ENCODING**, Ref. No.: RDI-0008US01; Commercialized at Rockfish Data, INC., `https://docs142.rockfish.ai/sdk/actions.html#rockfish.actions.StabPreprocess` and `https://docs142.rockfish.ai/sdk/actions.html#rockfish.actions.StabPostprocess`.	*Patent*
2025	**Provisional Patent Filing: Diffusion-Based Network Traffic Generation**, MBHB Ref. 24-1051-US-PRO	*Patent*
2025	**Non-Provisional Patent Filed: Diffusion-Based Network Traffic Generation**,	*Patent*
2025	**Provisional Patent Filing: Real-Time Recommendation and Planning Ecosystem for Physical Social Engagement**, commercialized at SOMI Social Corporation, `https://www.socialwithsome.com/`.	*Patent*

Honors & Awards

2022	**Best Presentation Award**, WIOTT at ICCSIT 2022	*Tokyo, Japan*
2021	**First Place Winner**, ACM Student Research Competition (SRC) at SIGCOMM	*Virtual (ACM)*
2021	**Crerar Fellowship**, the University of Chicago	*Chicago, IL*
2021	**Laura Sanchis Award for Excellence in Research**, Colgate University	*Hamilton, NY*
2021	**High Honors in Computer Science**, Colgate University CS Department	*Hamilton, NY*
2017-21	**Dean's Award with Distinction for Academic Excellence**, Colgate University	*Hamilton, NY*
2017	**Certificate of Recognition For Outstanding Artwork Selected for Display**, Chapman Cultural Center	*Spartanburg, SC*
2017	**Top Cooperation Award of Young Talent Program Social Practice Camp**, China Everbright Bank	*Shenyang, China*
2016	**Outstanding Intern Award**, Staubli Robotics	*Greenvile, SC*
2016	**Second Place Award in D1 Clemson Calculus Challenge**, Clemson University	*Clemson, SC*
2016	**Outstanding Delegate Award**, SC Model Arab League	*Clemson, SC*
2015	**Furman Scholar**, Furman University	*Greenvile, SC*
2015	**Wofford Scholar**, Wofford University	*Spartanburg, SC*

Academic Services

2026	**Reviewer (1 paper)**, IEEE Transactions on Network and Service Management (IEEE TNSM)	*Journal*
2025-Present	**Member**, NSF ACTION Institute Student Advisory Council (https://action.ucsb.edu/people)	
2025-2026	**TPC Member (17 papers)**, ACM Internet Measurement Conference (IMC)	*Karlsruhe, Germany*
2025	**Reviewer (2 papers)**, Engineering Applications of Artificial Intelligence	*Journal*
2025	**Reviewer (2 papers)**, Transactions on Mobile Computing	*Journal*
2025	**Reviewer (3 papers)**, IEEE Internet of Things Journal	*Journal*
2025	**Reviewer (3 papers)**, Computer Networks (COMNET)	*Journal*
2024-2025	**Reviewer (3 paper)**, IEEE/ACM Transaction on Networking	*Journal*
2024	**Reviewer (1 paper)**, Conference on Neural Information Processing Systems (NeurIPS)	*Vancouver, Canada*
2024	**TPC Member (6 papers)**, ACM Internet Measurement Conference (IMC)	*Madrid, Spain*
2021	**Proposal Writer**, Competition at the Genetic and Evolutionary Computation Conference (GECCO'21)	*Virtual*
2020-21	**Research Assistant**, Colgate University CS Department	*Hamilton, NY*

Media Coverage

University of Chicago Researchers Revolutionize Network Traffic Generation with AI Breakthrough — *UChicago CS News*

March 2025

`http://cs.uchicago.edu/news/university-of-chicago-researchers-revolutionize-network-traffic-generation-with-ai-breakthrough/`